



**CHAITANYA BHARATHI**  
**INSTITUTE OF TECHNOLOGY**  
Autonomous Institute | Affiliated to Osmania University

**FULL STACK WEB DEVELOPMENT (22CAC09N)**  
**CASE STUDY REPORT**  
**college Bus tracking & pass management**  
**system**

**College Bus Tracking & Pass Management**  
**System: A Web-Based Student**  
**Transportation Management Application**

**Submitted by:**

B.chandravardhan | RollNumber: 160124748096

D.Rajesh | Roll Number: 160124748103

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## College Bus Tracking & Pass Management System: A Web-Based Student Transportation Management Application

### 1. Introduction

College students often depend on campus buses for daily transportation, but information about bus routes, timings, available buses, and pass status is not always easy to access. Students may have to manually check schedules or contact transport staff to know which bus to take, whether a bus is available, or whether their bus pass has been approved. This creates the core problem addressed in this case study: how should a web application be designed so that students can easily view bus information, check routes and timings, apply for a bus pass, and track their pass status through a single interface?

This topic is important in full stack development because it involves user input handling and validation, data retrieval and separation of data from presentation, and responsive frontend state management. The important challenge is connecting user actions, application data, state, and UI updates into one coherent working flow. The same pattern can be applied to other transportation and campus-management applications. The proposed system provides students with a centralized platform to view bus routes, timings, available buses, driver information, and submit bus pass applications. The current version is designed as a frontend prototype using simulated JSON data, while a real backend and database can be added in future development.

### 2. Methods Compared

The College Bus Tracking & Pass Management System can be understood by comparing the data-source approach used in the current prototype with the approach planned for real deployment.

#### 2.1 Local JSON via Fetch API (Current) vs. Hospital API + Database (Planned)

- **Local JSON + Fetch API (current implementation):** Bus routes, timings, bus details, and pass information are stored in simulated JSON files. The React application retrieves this information asynchronously using the Fetch API (AJAX) and stores it in React state. This keeps the data separate from the UI and makes the prototype easy to build and modify, but the data is simulated rather than live transport data.
- **Bus Management API + Database (planned):** React would communicate with a backend API connected to a college transport database. This would allow the application to work with actual bus schedules, routes, bus availability, student pass applications, and driver information. Authentication and transport-admin functionality could also be added in a real deployment. The backend API and database are not currently implemented.

The two approaches share the same frontend pattern: because the React UI already loads information through the Fetch API, moving from local JSON to a real bus-management API mainly changes the data source rather than the interface design.



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## 2.2 Technology Stack

| Layer                 | Technology                    | Purpose                                                                   |
|-----------------------|-------------------------------|---------------------------------------------------------------------------|
| Structure             | HTML / JSX                    | Structure of the interface and React component markup                     |
| Styling               | CSS                           | Layout, styling, responsive design and dashboard styling                  |
| UI Framework          | Bootstrap 5                   | Responsive grid, buttons, forms, navbar, alerts, tables and UI components |
| Application Logic     | JavaScript                    | Input validation, event handling, filtering and pass-related calculations |
| Frontend UI           | React.js (Vite)               | Creating components, managing state, dynamic rendering and UI updates     |
| Data Retrieval        | Fetch API / AJAX              | Asynchronous retrieval of bus and route data                              |
| Data Source (Current) | Local JSON                    | Simulated bus routes, timings, buses, drivers and pass information        |
| Data Source (Planned) | Bus Management API + Database | Real-time bus information, route data and student pass records            |

## 3. Architecture

The College Bus Tracking & Pass Management System separates the application into three logical layers: a React frontend, a JavaScript logic and state layer, and a data layer that currently uses simulated JSON data.

**User → React UI → JavaScript Logic + State → Fetch API/AJAX → JSON Data → React UI Update**

### 3.1 High-Level Data Flow

- The user performs an action in the browser, such as selecting a bus route, checking bus timings, or submitting a bus pass application.
- React displays the interface and manages the application components.
- JavaScript performs input validation, filtering and pass-related calculations.
- React state stores changing information such as the selected route, bus details and pass application status.
- The Fetch API retrieves bus and route data asynchronously from the JSON file.
- The retrieved data is displayed dynamically through the React interface.

### 3.2 Application Workflow

The patient moves through the application in six steps:

- Home Page: The starting point for the student.
- Bus Routes: The student views available routes and route information.
- Bus Details: The student checks bus timings, availability and driver information.
- Pass Application: The student enters the required details and applies for a bus pass.
- Validate Details: The application checks that the required information has been entered correctly.
- Pass Status: The student views the submitted application and its current status.

**Home Page → Bus Routes → Bus Details → Pass Application → Validate Details  
→ Pass Status**



### 3.3 State Update Logic (Frontend)

- On load, the application fetches data.json and stores the bus route and bus information in React state.
- The selected route, selected bus and pass application status are held in React state.
- When the selected route or pass status changes, React automatically re-renders the relevant part of the interface without requiring a manual page refresh

selectedRoute → setSelectedRoute() → React re-renders → Bus

## 4. Implementation

Selected implementation excerpts illustrating the architecture described above:

### 4.1 Frontend — Fetch API: Load Department Data (src/App.jsx)

```
</> JavaScript

useEffect(() => {
  fetch("/data.json")
    .then(res => res.json())
    .then(data => {
      setBuses(data.buses);
      setRoutes(data.routes);
    });
}, []);
```

Flow:

fetch data.json → convert response to JSON → get buses and routes → store data in React state

This avoids hard-coding bus and route details directly in the UI.

### 4.2 Frontend — React State: Track Bus & Pass Status (src/App.jsx)

```
</> JavaScript

const [selectedRoute, setSelectedRoute] = useState(null);
const [selectedBus, setSelectedBus] = useState(null);
const [passStatus, setPassStatus] = useState("Not Applied");
```

selectedRoute stores the route selected by the student, selectedBus stores the selected bus information, and passStatus stores the current bus-pass application status.

When any of these values changes, React re-renders the relevant part of the interface.

### 4.3 JavaScript — Filter Available Buses

```
</> JavaScript

const availableBuses = buses.filter(
  bus => bus.routeId === selectedRoute
);
```



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The application filters buses according to the route selected by the student.  
For example, if a route has 3 available buses, only those buses are displayed for that selected route.

## 4.4 Project Structure and Key Files

```
college-bus/  
├── public/  
│   └── data.json  
├── src/  
│   ├── App.jsx  
│   ├── App.css  
│   └── main.jsx  
├── index.html  
├── package.json  
└── vite.config.js
```

## 4.4 Project Structure and Key Files

| File           | Purpose                                                      |
|----------------|--------------------------------------------------------------|
| App.jsx        | Main React application, components, state and bus/pass logic |
| App.css        | Styling, layout and responsive design                        |
| main.jsx       | Entry point of the React application                         |
| data.json      | Simulated bus, route, driver and pass information            |
| package.json   | Project dependencies and configuration                       |
| vite.config.js | Vite development configuration                               |

## 5. Results

The implemented prototype was evaluated qualitatively against its functional goals rather than through a formal load test. The main problems addressed during development, and how each was resolved, were:

**Dynamic bus information:** Bus and route information is loaded from the JSON data source using the Fetch API and displayed dynamically through React state.

**Separating data from the UI:** Route, bus and driver information is stored in data.json, fetched using the Fetch API and held in React state before being displayed in the interface. This allows the data to be changed without modifying the UI.

**Route-based bus filtering:** JavaScript filters the available buses according to the route selected by the student, allowing relevant bus information to be displayed.

**Pass application and validation:** The pass application form validates the required student information before submitting the application and displaying the pass status.

## 6. Conclusion

This case study demonstrates that building a full stack web application involves more than creating a user interface: the key is connecting user actions, application data, state and UI updates into one complete and working flow. The College Bus Tracking & Pass Management System demonstrates this approach at prototype level: React state manages changing bus and pass information, the Fetch API separates data from the interface, and JavaScript handles validation, filtering and application logic. The current system is primarily a frontend prototype using simulated JSON data.

Future work includes:

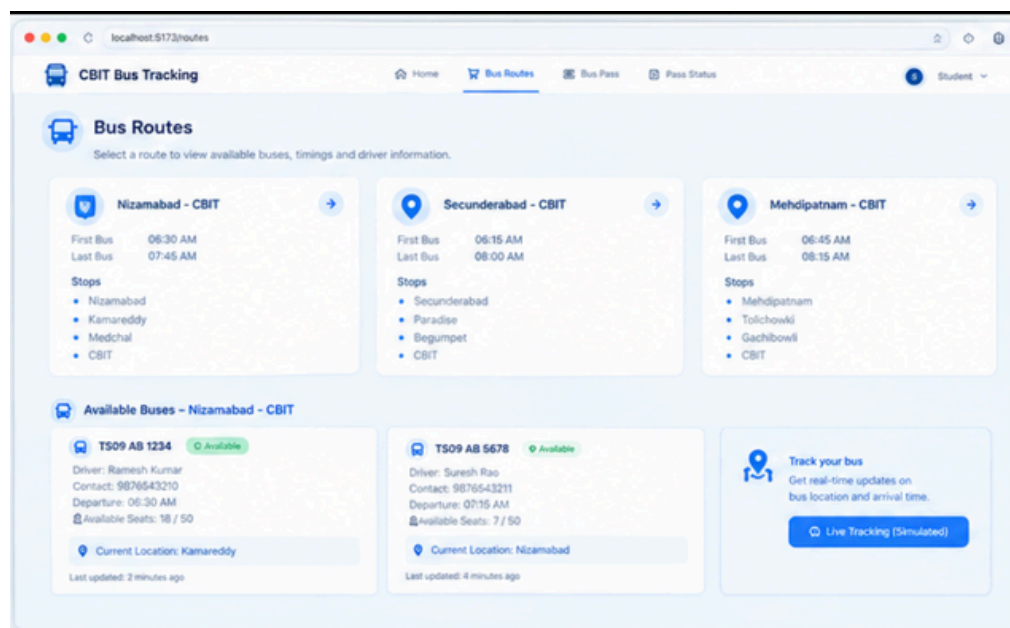
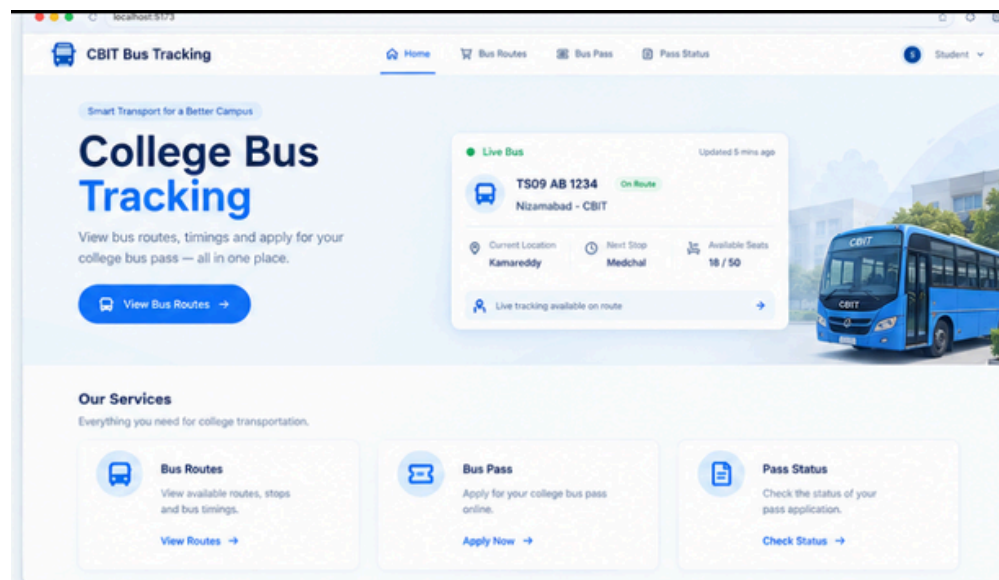
- (1) a real backend and database so the application can work with actual college transport information;
- (2) integration with GPS/location services for real-time bus tracking;
- (3) a transport-admin dashboard for managing buses, routes, drivers and pass applications;
- (4) authentication for students and transport staff; and
- (5) notifications to inform students about bus arrival, route changes and pass approval.



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OUTPUT:





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OUTPUT:

localhost:5173/pass

CBIT Bus Tracking

Home Bus Routes **Bus Pass** Pass Status Student

### Bus Pass Application

Enter your details to apply for a college bus pass.

**Student Name \***  
Enter your name

**Roll Number \***  
Enter roll number

**Mobile Number \***  
1234567890

Please enter a valid 10-digit mobile number.

**Select Route \***  
Select a route

Submit Application

### Pass Status

Check the status of your bus pass application.

**Application #BP2025001**

|             |                       |
|-------------|-----------------------|
| Name        | Chandravardhan        |
| Roll Number | 22CS123               |
| Route       | Nizamabad - CBIT      |
| Applied On  | 22 Sep 2025, 10:30 AM |

Pending

Your application is under review. You will be notified once the status is updated.



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